Practical Exam Master Guide (Q1 to Q44)

Q1. Import a CSV file into Tableau and display the imported data in a worksheet using Bar Chart.

- **i)** Connect an Excel file to Tableau and verify the detected data types
- **ii)** Classify the imported fields as Dimensions and Measures.

Answer:

- **Step-by-Step Guide:**
  1. Open Tableau Desktop → Click **Connect** → **Microsoft Excel** (or **Text File** for CSV).
  2. Select `StudentData.xlsx` or `StudentData.csv`.
  3. In the Data Source screen, verify the auto-detected data types above each field header:
     - `StudentID`: String (Abc)
     - `Name`: String (Abc)
     - `Department`: String (Abc)
     - `Marks`: Whole Number (#)
  4. Navigate to **Sheet 1**.
  5. **Classification Verification:**
     - **Dimensions (Categorical/Blue):** `StudentID`, `Name`, `Department`
     - **Measures (Numerical/Green):** `Marks`
  6. **Create Bar Chart:** Drag `Name` to **Columns** shelf, drag `Marks` to **Rows** shelf. Tableau will render a Bar Chart automatically.

Q2. Create a dataset of 10 students with columns Student ID, Name, Department and Marks in MS Excel and save it as StudentData.xlsx or .csv. Identify the data type of each field and complete the table below. Classify the imported fields as Dimensions and Measures.

Field Name	Data Type
StudentID	
Name	
Department	
Marks	

Answer:

- **Completed Table:**

Field Name	Data Type	Tableau Classification
`StudentID`	String / Text	Dimension
`Name`	String / Text	Dimension
`Department`	String / Text	Dimension
`Marks`	Whole Number	Measure

- **Steps:** Save the table in Excel as `StudentData.xlsx`. Open Tableau, connect to this file, and confirm that `StudentID`, `Name`, and `Department` are categorized under Dimensions (Blue) and `Marks` is categorized under Measures (Green).

Q3. Create a dataset in Excel, save it as a csv file containing ProductID, ProductName, Category, and Price. Import it into Tableau and classify fields as Dimensions and Measures.

Answer:

- **Data Classification:**
  - `ProductID`: Dimension (String or Number Discrete)
  - `ProductName`: Dimension (String)
  - `Category`: Dimension (String)
  - `Price`: Measure (Continuous Decimal / Number)
- **Step-by-Step Guide:**
  1. Create a table in Excel with headers `ProductID`, `ProductName`, `Category`, `Price`.
  2. Go to **File** → **Save As** → File Format: **CSV (Comma delimited) (*.csv)**.
  3. In Tableau, select **To a File** → **Text File** → Choose your saved CSV.
  4. In **Sheet 1**, check the Data Pane:
     - Move `ProductID`, `ProductName`, and `Category` to **Dimensions** (top pane).
     - Move `Price` to **Measures** (bottom pane).

Q4. Enter the following data into a txt file:

ProductName	Price
Laptop	55000
Mobile	25000
Tablet	18000
Printer	12000
Monitor	15000
Keyboard	1200

- **i)** Import the text (.txt) file into Tableau and verify the imported records.
- **ii)** Construct a Bar Chart (Product vs Price) in Tableau.
- **iii)** Construct a Line Chart (Product vs Price) in Tableau.
- **iv)** Create an interactive Dashboard in Tableau using the Bar Chart and Line Chart. Format the dashboard by adding a title, legends, and suitable layout components.

Answer:

- **Step-by-Step Solution:**
  1. Save data into `ProductData.txt` using Tab or Comma delimiters.
  2. **Import:** Connect to **Text file** in Tableau → Select `ProductData.txt`.
  3. **Worksheet 1 (Bar Chart):** Drag `ProductName` to **Columns**, `Price` to **Rows**. Title sheet `Product Bar Chart`.
  4. **Worksheet 2 (Line Chart):** Drag `ProductName` to **Columns**, `Price` to **Rows**. In Marks Card, change dropdown from **Automatic** to **Line**. Title sheet `Product Line Chart`.
  5. **Interactive Dashboard:**
     - Click **New Dashboard**. Set Size to **Automatic**.
     - Drag `Product Bar Chart` and `Product Line Chart` onto the canvas side-by-side.
     - Click **Objects** → **Text** → Add Header: `"Product Price Analysis Dashboard"`.
     - Select the Bar Chart → Click the **Use as Filter** funnel icon to link interaction across both charts.

Q5. Creating a Sales Analysis Dashboard using Tableau:

- **i)** Brand-wise Units Sold (Bar Chart)
- **ii)** Quarter-wise Revenue (Pie Chart)
- **iii)** Brand wise Revenue (Bubble Chart)
- **iv)** Brand-wise Units Sold (Pie Chart)

Answer:

- **Step-by-Step Solution:**
  1. **Sheet 1 (Bar Chart):** `Brand` to **Columns**, `Units Sold` to **Rows**. Select Bar Chart.
  2. **Sheet 2 (Pie Chart):** Mark type = **Pie**. Drag `Quarter` to **Color**, `Revenue` to **Angle**.
  3. **Sheet 3 (Bubble Chart):** Mark type = **Circle** (or select Packed Bubbles from *Show Me*). Drag `Brand` to **Detail**, `Revenue` to **Size** and **Color**.
  4. **Sheet 4 (Pie Chart):** Mark type = **Pie**. Drag `Brand` to **Color**, `Units Sold` to **Angle**.
  5. **Dashboard:** Combine all 4 views into a 2 × 2 grid container. Add Title and Legends.

Q6. Import the Online Shopping dataset into Tableau:

- **i)** Create a Pie Chart showing Category-wise Total Amount.
- **ii)** Create a Bar Chart showing distribution of orders according to Payment Mode.
- **iii)** Create a Sorted Horizontal Bar Chart showing Region-wise Total Amount.
- **iv)** Create a Tree Map showing Category-wise Amount.
- **v)** Create a Dashboard containing any three visualizations created above with proper formatting.

Answer:

- **Step-by-Step Solution:**
  1. **Pie Chart:** `Category` → **Color**, `Amount` → **Angle**.
  2. **Bar Chart:** `Payment Mode` → **Columns**, `Order ID` (Count) → **Rows**.
  3. **Sorted Horizontal Bar Chart:** `Total Amount` → **Columns**, `Region` → **Rows**. Click the **Sort Descending** toolbar button.
  4. **Tree Map:** `Category` → **Label**, `Amount` → **Size** & **Color**. Select **Treemap** from Show Me.
  5. **Dashboard:** Drag 3 sheets onto the Dashboard workspace, format borders, colors, and layout containers.

Q7. Import the College Placement dataset into Tableau:

- **i)** Create a Pie Chart showing Department-wise Average Package (LPA).
- **ii)** Create an Appropriate Chart showing Gender-wise Student Distribution.
- **iii)** Create a Bar Chart showing Company-wise Number of Placements.
- **iv)** Create a Highlight Table showing Company and Package Filtered by Department.
- **v)** Create an Interactive Dashboard containing the visualizations created above and format it appropriately using titles, colors, sorting, and legends.

Answer:

- **Step-by-Step Solution:**
  1. **Pie Chart:** `Department` → **Color**, `Package (LPA)` (Change aggregation to **Average**) → **Angle**.
  2. **Gender Distribution:** `Gender` → **Columns**, `StudentID` (Count) → **Rows** (Bar chart).
  3. **Company Placements:** `Company` → **Columns**, `StudentID` (Count) → **Rows**.

4. **Highlight Table:** Place `Company` on **Rows**, `Department` on **Columns**. Drag `Package` to **Text** and **Color** cards.
5. **Dashboard:** Combine sheets, add interactive filters for `Department`.

Q8. Create a story with the two dashboards created in Set A and Set B.

Answer:

- **Step-by-Step Solution:**
  1. In Tableau, click **New Story** tab (bottom menu next to Dashboard tab).
  2. In the left Story pane, drag **Dashboard 1** into the story point canvas.
  3. Add a descriptive caption header at the top (e.g., `"Executive Summary: Sales Overview"`).
  4. Click **Blank** in the left pane to add a second story point page.
  5. Drag **Dashboard 2** into the second page canvas and add caption (e.g., `"Detailed Placement Analysis"`).

Q9. Download any appropriate dataset from Kaggle and create an interactive data visualization dashboard.

Answer:

- **Step-by-Step Solution:**
  1. Go to kaggle.com, download any dataset (e.g., `Superstore.csv`).
  2. Open Tableau → Connect to `Superstore.csv`.
  3. Create Visuals:
     - Worksheet 1: `Sales` by `Sub-Category` (Horizontal Bar Chart).
     - Worksheet 2: `Profit` over `Order Date` (Line Chart).
     - Worksheet 3: `Sales` by `State` (Map).
  4. Assemble all 3 worksheets into a Dashboard and enable action filters.

Q10. Import Superstore Sales Dataset into Tableau:

- **i)** Connect the dataset to Tableau.
- **ii)** Create a bar chart showing Sales by Category.
- **iii)** Create a line chart showing Sales over time.
- **iv)** Add a filter for Region.
- **v)** Create a dashboard combining both visualizations. Add a suitable dashboard title.

Answer:

- **Step-by-Step Solution:**
  1. Connect Tableau to `Sample - Superstore.xls`.
  2. **Bar Chart:** Drag `Category` to **Columns**, `Sales` to **Rows**.
  3. **Line Chart:** Drag `Order Date` (Year/Month) to **Columns**, `Sales` to **Rows**.
  4. **Filter:** Drag `Region` to **Filters** shelf. Right-click filter card → **Apply to Worksheets** → **All Using This Data Source**.
  5. **Dashboard:** Combine Bar and Line charts, set top header text box to `"Superstore Performance Analysis"`.

Q11. Import Sales Dataset into Tableau:

- **i)** Create a visualization comparing Sales and Profit by Category.
- **ii)** Use a suitable chart such as a dual-axis chart.
- **iii)** Format the axes appropriately.

- **iv)** Add a Category filter.
- **v)** Create a dashboard containing the visualization.

Answer:

- **Step-by-Step Solution:**
  1. Drag `Category` to **Columns**.
  2. Drag `Sales` to **Rows**. Drag `Profit` to **Rows** (next to Sales).
  3. Right-click `SUM(Profit)` pill on Rows → Select **Dual Axis**.
  4. Right-click the right-side vertical axis → Click **Synchronize Axis**.
  5. Drag `Category` to **Filters** shelf.
  6. Place sheet onto a new Dashboard with title formatting.

Q12. Create three worksheets: (1) Sales by Category, (2) Profit by Region, (3) Sales Trend. Create a dashboard using all three worksheets. Apply appropriate dashboard formatting. Add a suitable dashboard title.

Answer:

- **Step-by-Step Solution:**
  1. **Sheet 1:** `Category` (Columns), `Sales` (Rows) → Bar Chart.
  2. **Sheet 2:** `Region` (Columns), `Profit` (Rows) → Column Chart.
  3. **Sheet 3:** `Order Date` (Columns), `Sales` (Rows) → Line Chart.
  4. **Dashboard:** Drag all three worksheets onto canvas. Format using tiled containers, color palettes, and title `"Executive Sales & Profit Metrics"`.

Q13. Power BI - Create the dataset in Excel and Import it into Power BI Desktop and perform the following data cleaning operations using Power Query Editor:

EmployeeID	Employee Name	Department	Salary	City
E101	Amit	HR	35000	Pune
E102	Sneha	IT	45000	Mumbai
E103	Rahul	Sales	40000	Pune
E104	Priya	IT		Nashik
E105	Karan	HR	35000	Pune
E105	Karan	HR	35000	Pune
E106	Neha	Finance	52000	Mumbai
E107	Rohan	Sales	41000	
E108	Aditi	IT	48000	Nagpur
E109	Vivek	Finance	50000	Mumbai

- **i)** Rename the column Employee Name as EmployeeName. (remove space)
- **ii)** Verify and correct the data type of each column.
- **iii)** Identify and remove duplicate records.
- **iv)** Identify rows containing missing (Blank/Null) values.
- **v)** Remove rows containing missing values.
- **vi)** Apply the changes and load the cleaned dataset into Power BI.

Answer:

- **Step-by-Step Solution in Power Query:**
  1. Get Data → Excel → Select file → Click **Transform Data**.
  2. **Rename Column:** Double-click header `Employee Name` → Change to `EmployeeName`.
  3. **Data Types:** Ensure `EmployeeID`, `EmployeeName`, `Department`, `City` are **Text**, and `Salary` is **Whole Number**.
  4. **Remove Duplicates:** Select all columns (Ctrl + A) → Right-click column header → **Remove Duplicates** (removes redundant `E105` row).
  5. **Remove Blank/Null Rows:**
     - Click drop-down filter on `Salary` column header → Uncheck **null**.
     - Click drop-down filter on `City` column header → Uncheck **null**.
  6. **Load:** Click **Home** → **Close & Apply**.

Q14. Dataset: AI Hardware Sales

TID	Product	Quantity	UnitPrice	ShippingCost	Discount
T101	NVIDIA RTX 5090 GPU	2	245000	1500	5000
T102	Apple MacBook Pro M4	1	289000	1000	8000
T103	AMD Ryzen 9 9950X	3	62000	800	3000
T104	Samsung 2TB NVMe SSD	4	18500	500	1200
T105	Dell AI Workstation	1	365000	2500	10000

Create the above dataset in MS Excel and save it as AIHardwareSales.xlsx. Import the dataset into Power BI Desktop and perform the following operations using Power Query Editor:

- **i)** Verify and correct the data type of each field.
- **ii)** Rename the column UnitPrice as PricePerUnit.
- **iii)** Create a new Custom Column named TotalAmount using the following formula:
  TotalAmount = (Quantity × PricePerUnit) + ShippingCost − Discount
- **iv)** Sort the records in descending order of TotalAmount.
- **v)** Apply the changes and load the transformed dataset into Power BI.

Answer:

- **Step-by-Step Solution:**
  1. Import `AIHardwareSales.xlsx` into Power Query Editor.
  2. Verify Data types: `TID`, `Product` (Text); `Quantity`, `UnitPrice`, `ShippingCost`, `Discount` (Whole/Decimal Numbers).
  3. Rename column `UnitPrice` to `PricePerUnit`.
  4. Click **Add Column** tab → **Custom Column**.
     - Column Name: `TotalAmount`
     - Custom Column Formula:

     ```
     ([Quantity] * [PricePerUnit]) + [ShippingCost] - [Discount]
     ```

  5. Click arrow on `TotalAmount` header → **Sort Descending**.
  6. Click **Home** → **Close & Apply**.

Q15. Create an Excel dataset containing StudentName, Department, and CGPA. Import it into Power BI and perform the following:

- **i)** Change the data type of CGPA to Decimal Number.

- **ii)** Sort the data by CGPA in descending order.
- **iii)** Filter the records to display only one department.
- **iv)** Load the transformed data into Power BI.

Answer:

- **Step-by-Step Solution:**
  1. Import file into Power Query Editor.
  2. Click data type icon next to `CGPA` header → Change to **Decimal Number**.
  3. Click arrow on `CGPA` header → **Sort Descending**.
  4. Click arrow on `Department` header → Uncheck all, select single department (e.g., `"Computer Science"`).
  5. Click **Close & Apply**.

Q16. Download any suitable dataset from Kaggle and import it into Power BI Desktop. Perform appropriate data cleaning operations on the dataset.

Answer:

- **Step-by-Step Solution:**
  1. Download CSV file from Kaggle.
  2. In Power BI: **Get Data** → **Text/CSV**.
  3. In Power Query Editor:
     - Click **Use First Row as Headers**.
     - Select columns with empty rows → **Remove Blank Rows**.
     - Select ID columns → **Remove Duplicates**.
  4. Click **Close & Apply**.

Q17. Create the following two Excel datasets viz. Department.xlsx and Student.xlsx.

DepartmentID	DepartmentName
D01	Data Science
D02	Computer Science
D03	Statistics

StudentID	StudentName	DepartmentID
101	Aarav	D01
102	Ishita	D02
103	Rohan	D03
104	Riya	D01
105	Karan	D02

- **i)** Import both datasets into Power BI.
- **ii)** Create a relationship between the two tables using DepartmentID.
- **iii)** Identify the relationship cardinality.

Answer:

- **Step-by-Step Solution:**
  1. Import both Excel files into Power BI Desktop (**Get Data** → **Excel workbook**).
  2. Click on **Model View** icon on the far left navigation bar (3rd icon down).

3. Left-click and drag `DepartmentID` from the `Department` table block and drop it directly onto `DepartmentID` in the `Student` table block.
4. **Relationship Cardinality Identification:**
   - Double-click the connecting line to verify parameters.
   - **Cardinality: One-to-Many (1 : ∗)** (`Department` is 1, `Student` is ∗).

Q18. Create the following two Excel datasets viz. Customer.xlsx and Orders.xlsx.

CustomerID	CustomerName	City
C101	Rahul	Pune
C102	Sneha	Mumbai
C103	Aman	Nagpur

OrderID	CustomerID	Amount
O101	C101	4500
O102	C101	3800
O103	C102	5200
O104	C103	2100
O105	C102	6700

- **i)** Import both datasets into Power BI.
- **ii)** Create a relationship using CustomerID.
- **iii)** Identify the relationship cardinality.
- **iv)** Display the data in Model View.

Answer:

- **Step-by-Step Solution:**
  1. Import both tables into Power BI.
  2. Click **Model View** icon on the left bar.
  3. Drag `CustomerID` from `Customer` table and drop onto `CustomerID` in `Orders` table.
  4. **Cardinality: One-to-Many (1 : ∗)** (Each customer can place multiple orders).

Q19. Import all four datasets into Power BI Desktop and perform data cleaning as required. Create relationships between the tables using the appropriate keys and Identify the relationship cardinality.

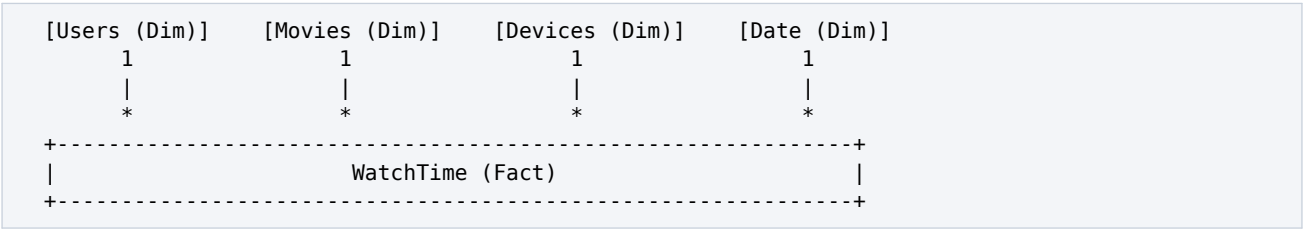
Answer:

- **Step-by-Step Solution:**
  1. Import `Users`, `Movies`, `Devices`, and `WatchTime` tables into Power BI.
  2. Click **Transform Data** in Home tab to clear empty rows and verify data types.
  3. Click **Model View** on the left menu.
  4. Drag primary keys to foreign keys to create relationships:
     - Drag `Users[UserID]` → Drop on `WatchTime[UserID]` (1 : ∗)
     - Drag `Movies[MovieID]` → Drop on `WatchTime[MovieID]` (1 : ∗)
     - Drag `Devices[DeviceID]` → Drop on `WatchTime[DeviceID]` (1 : ∗)
     - Drag `Date[DateID]` → Drop on `WatchTime[DateID]` (1 : ∗)

Q20. Build a Star Schema with WatchTime as Fact Table. Identify the Fact Table and the Dimension Tables. Display the completed data model in Model View.

Answer:

- **Schema Identification:**
  - **Fact Table:** `WatchTime` (Contains numeric measures like watch duration and keys).
  - **Dimension Tables:** `Users`, `Movies`, `Devices`, `Date` (Contain descriptive attributes).
- **Click-by-Click Steps to Build Star Schema in Power BI:**
  1. Click the **Model View** icon (3rd icon on the left sidebar).
  2. Arrange table blocks on the canvas: Drag the `WatchTime` block to the center. Place `Users`, `Movies`, `Devices`, and `Date` surrounding `WatchTime`.
  3. **Establish Relationships:**
     - Click and drag `Users[UserID]` → Drop on `WatchTime[UserID]`.
     - Click and drag `Movies[MovieID]` → Drop on `WatchTime[MovieID]`.
     - Click and drag `Devices[DeviceID]` → Drop on `WatchTime[DeviceID]`.
     - Click and drag `Date[DateID]` → Drop on `WatchTime[DateID]`.
  4. Double-click each relationship line to open the **Edit Relationship** window:
     - Verify **Cardinality** is set to `One to Many (1:*)`.
     - Verify **Cross filter direction** is set to `Single`.
     - Click **OK**.



Q21. Online MOOC Courses Dataset: Import the dataset into Power BI and create DAX Measures for Total Course Revenue using SUM(), Average Course Fee using AVERAGE(), Maximum Course Fee using MAX(), Minimum Course Fee using MIN(), Total Enrollments using COUNT().

Answer:

- **DAX Formulas:**

```
Total Course Revenue = SUM(Courses[Course Fee])
```

```
Average Course Fee = AVERAGE(Courses[Course Fee])
```

```
Maximum Course Fee = MAX(Courses[Course Fee])
```

```
Minimum Course Fee = MIN(Courses[Course Fee])
```

```
Total Enrollments = COUNT(Courses[EnrollmentID])
```

Q22. Create the calculated column Course Category using IF(): Course Fee >= ₹7000 -> Premium, Otherwise -> Standard.

Answer:

- **DAX Formula (Calculated Column):**

```
Course Category =
IF(
    Courses[Course Fee] >= 7000,
    "Premium",
    "Standard"
)
```

- **Step:** Go to **Data View** → Click **New Column** → Enter formula above.

Q23. Create the DAX Measures using Time Intelligence functions: Year-to-Date Revenue using TOTALYTD(), Month-to-Date Revenue using TOTALMTD(), Quarter-to-Date Revenue using TOTALQTD().

Answer:

- **DAX Formulas:**

```
Year-to-Date Revenue = TOTALYTD(SUM(Courses[Course Fee]), 'Date'[Date])
```

```
Month-to-Date Revenue = TOTALMTD(SUM(Courses[Course Fee]), 'Date'[Date])
```

```
Quarter-to-Date Revenue = TOTALQTD(SUM(Courses[Course Fee]), 'Date'[Date])
```

Q24. Create an interactive report containing Card showing Total Course Revenue, Card showing Average Course Fee, Line Chart showing Monthly Average Revenue by Enrollment Date, Slicers for Course and Instructor.

Answer:

- **Step-by-Step Solution:**
  1. Add **Card 1** visual → Drag `Total Course Revenue` measure.
  2. Add **Card 2** visual → Drag `Average Course Fee` measure.
  3. Add **Line Chart** → X-axis: `Enrollment Date` (Month), Y-axis: `Average Course Fee`.
  4. Add **Slicer 1** → Drag `Course` field.
  5. Add **Slicer 2** → Drag `Instructor` field.

Q25. Iris Dataset: Import the Iris Dataset into Power BI and create DAX Measures for Average Sepal Length using AVERAGE(), Maximum Petal Length using MAX(), Minimum Sepal Width using MIN(), Total Flowers using COUNT().

Answer:

- **DAX Formulas:**

```
Average Sepal Length = AVERAGE(Iris[SepalLength])
```

```
Maximum Petal Length = MAX(Iris[PetalLength])
```

```
Minimum Sepal Width = MIN(Iris[SepalWidth])
```

```
Total Flowers = COUNT(Iris[Species])
```

Q26. Add a slicer based on variety and find variety wise column averages.

Answer:

- **Step-by-Step Solution:**
  1. Drag a **Slicer** visual to canvas → Add `Variety` field.
  2. Create a **Table Visual** → Add `Variety`, `Average Sepal Length`, `Average Sepal Width`, `Average Petal Length`, `Average Petal Width`.
  3. Selecting values in the Variety slicer filters average metrics automatically.

Q27. Create a Calculated Column named Species Code using the SWITCH() function with mapping: Setosa -> 1, Versicolor -> 2, Virginica -> 3.

Answer:

- **DAX Calculated Column:**

```
Species Code =
SWITCH(
    Iris[Variety],
    "Setosa", 1,
    "Versicolor", 2,
    "Virginica", 3,
    BLANK()
)
```

Q28. Create an interactive report containing Cards and Scatter Plots for the Iris dataset.

Answer:

- **Step-by-Step Solution:**
  1. **Card 1:** Add `Average Sepal Length`.
  2. **Card 2:** Add `Maximum Petal Length`.
  3. **Scatter Plot 1:** X-axis = `SepalLength`, Y-axis = `SepalWidth`, Legend = `Variety`.
  4. **Scatter Plot 2:** X-axis = `PetalLength`, Y-axis = `PetalWidth`, Legend = `Variety`.

Q29. Create a bar chart showing the marks of 5 students using Python in Power BI.

Answer:

- **Where to Write and Execute the Code in Power BI:**
  1. In Power BI Desktop **Report View**, click the **Py** (Python visual) icon in the **Visualizations** pane.
  2. Drag `StudentName` and `Marks` from the **Data** pane into the **Values** field bucket of the Python visual.
  3. The **Python script editor** pane will pop up at the bottom of the screen.
  4. Copy and paste the script below into the editor, then click the **Run** (Play) button at the top-right of the script editor window.
- **Python Code Script:**

```
import matplotlib.pyplot as plt

plt.figure(figsize=(6, 4))
```

```
plt.bar(dataset['StudentName'], dataset['Marks'], color='skyblue')
plt.xlabel('Student Name')
plt.ylabel('Marks')
plt.title('Marks of 5 Students')
plt.show()
```

Q30. Draw a histogram for the ages of ten employees using Python in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Click **Py** icon in **Visualizations** pane.
  2. Drag `Age` field into the **Values** field section.
  3. Paste the code into the bottom **Python script editor** and click the **Run** button.
- **Python Code Script:**

```
import matplotlib.pyplot as plt

plt.figure(figsize=(6, 4))
plt.hist(dataset['Age'], bins=5, color='green', edgecolor='black')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Employee Age Distribution')
plt.show()
```

Q31. Draw a pie chart showing the distribution of sales among five products using Python in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Click **Py** visual icon in **Visualizations** pane.
  2. Drag `Product` and `Sales` fields into **Values**.
  3. Paste code into **Python script editor** and click **Run**.
- **Python Code Script:**

```
import matplotlib.pyplot as plt

plt.figure(figsize=(5, 5))
plt.pie(dataset['Sales'], labels=dataset['Product'], autopct='%1.1f%%', startangle=90)
plt.title('Sales Distribution')
plt.show()
```

Q32. Create a Line Chart showing monthly sales for six months using Python in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Add **Py** visual to report.
  2. Drag `Month` and `Sales` fields into **Values**.
  3. Paste code into **Python script editor** and click **Run**.
- **Python Code Script:**

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(6, 4))
plt.plot(dataset['Month'], dataset['Sales'], marker='o', color='red', linestyle='-')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Trend')
plt.grid(True)
plt.show()
```

Q33. Draw a Box Plot showing the marks of ten students using Python in Power BI. Also calculate the descriptive statistics and display it using Python.

Answer:

- **Where to Write and Execute Code:**
  1. Add **Py** visual to canvas.
  2. Drag `Marks` into **Values**.
  3. Paste code in **Python script editor** and click **Run**.
- **Python Code Script:**

```
import matplotlib.pyplot as plt

# Print descriptive statistics in output console
print(dataset['Marks'].describe())

plt.figure(figsize=(5, 4))
plt.boxplot(dataset['Marks'], vert=True, patch_artist=True)
plt.ylabel('Marks')
plt.title('Student Marks Box Plot')
plt.show()
```

Q34. Draw a Scatter Plot showing the relationship between Hours Worked and Income using Python in Power BI. Fit a Simple Linear Regression model and display the regression line.

Hours Worked	Income (₹)
2	500
3	970
4	900
5	1100
6	1300
7	1000
8	1700
9	1900
10	2100
11	2300

Answer:

- **Where to Write and Execute Code:**
  1. Click **Py** visual icon in **Visualizations** pane.
  2. Drag `Hours Worked` and `Income` fields into **Values**.

3. Paste code into **Python script editor** and click **Run**.

- **Python Code Script:**

```
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

X = dataset[['Hours Worked']].values
y = dataset['Income'].values

model = LinearRegression()
model.fit(X, y)

plt.figure(figsize=(6, 4))
plt.scatter(X, y, color='blue', label='Data Points')
plt.plot(X, model.predict(X), color='red', linewidth=2, label='Regression Line')
plt.xlabel('Hours Worked')
plt.ylabel('Income (₹)')
plt.title('Hours Worked vs Income')
plt.legend()
plt.show()
```

Q35. Predict Salary Based on Years of Experience Using Simple Linear Regression in Python in Power BI. Estimate the regression Equation and calculate summary statistics.

Experience (Years)	Salary (₹ Thousand)
1	25
2	30
3	36
4	42
5	75
6	55
7	85
8	98
9	125
10	130

Answer:

- **Where to Write and Execute Code:**
  1. Add **Py** visual to report canvas.
  2. Drag `Experience` and `Salary` into **Values**.
  3. Paste code into **Python script editor** and click **Run**.
- **Python Code Script:**

```
import pandas as pd
from sklearn.linear_model import LinearRegression

X = dataset[['Experience']].values
y = dataset['Salary'].values

model = LinearRegression()
model.fit(X, y)

print("Summary Statistics:")
```

```
print(pd.DataFrame({'Experience': X.flatten(), 'Salary': y}).describe())
print(f"\nRegression Equation: Salary = {model.intercept_:.2f} + {model.coef_[0]:.2f} *
    Experience")
```

Q36. Create a bar chart showing the marks of 5 students using R in Power BI.

Answer:

- **Where to Write and Execute the Code in Power BI:**
  1. In Power BI Desktop **Report View**, click the **R** (R visual) icon in the **Visualizations** pane.
  2. Drag `StudentName` and `Marks` from the **Data** pane into the **Values** field bucket of the R visual.
  3. The **R script editor** pane will pop up at the bottom of the screen.
  4. Copy and paste the script below into the editor, then click the **Run** (Play) button at the top-right of the script editor window.
- **R Code Script:**

```
library(ggplot2)

ggplot(dataset, aes(x = StudentName, y = Marks)) +
  geom_bar(stat = "identity", fill = "cornflowerblue") +
  theme_minimal() +
  labs(title = "Student Marks", x = "Student", y = "Marks")
```

Q37. Draw a histogram for the ages of ten employees using R in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Click **R** icon in **Visualizations** pane.
  2. Drag `Age` field into **Values**.
  3. Paste code into **R script editor** at the bottom and click **Run**.
- **R Code Script:**

```
hist(dataset$Age,
     main = "Employee Age Distribution",
     xlab = "Age",
     col = "lightgreen",
     border = "black")
```

Q38. Draw a pie chart showing the distribution of sales among five products using R in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Click **R** visual icon in **Visualizations** pane.
  2. Drag `Product` and `Sales` fields into **Values**.
  3. Paste code into **R script editor** and click **Run**.
- **R Code Script:**

```
slices <- dataset$Sales
lbls <- dataset$Product
pct <- round(slices/sum(slices)*100)
```

```
lbls <- paste(lbls, pct, "%", sep=" ")
pie(slices, labels = lbls, main = "Sales Distribution", col = rainbow(length(lbls)))
```

Q39. Create a Line Chart showing monthly sales for six months using R in Power BI.

Answer:

- **Where to Write and Execute Code:**
  1. Add **R** visual to report.
  2. Drag `Month` and `Sales` fields into **Values**.
  3. Paste code into **R script editor** and click **Run**.
- **R Code Script:**

```
plot(dataset$Month, dataset$Sales, type = "o", col = "blue",
     main = "Monthly Sales Trend", xlab = "Month", ylab = "Sales", pch = 16)
```

Q40. Draw a Box Plot showing the marks of ten students using R in Power BI. Also calculate the descriptive statistics and display it using R.

Answer:

- **Where to Write and Execute Code:**
  1. Add **R** visual to canvas.
  2. Drag `Marks` into **Values**.
  3. Paste code in **R script editor** and click **Run**.
- **R Code Script:**

```
print("Descriptive Statistics:")
print(summary(dataset$Marks))

boxplot(dataset$Marks, main = "Box Plot of Student Marks",
        ylab = "Marks", col = "orange")
```

Q41. Draw a Scatter Plot showing the relationship between Study Hours and Marks using R in Power BI. Fit a Simple Linear Regression model and display the regression line.

Study Hours	Marks
1	35
2	42
3	55
4	50
5	65
6	68
7	75
8	78
9	90
10	85

Answer:

- **Where to Write and Execute Code:**
  1. Click **R** visual icon in **Visualizations** pane.
  2. Drag `StudyHours` and `Marks` fields into **Values**.
  3. Paste code into **R script editor** and click **Run**.
- **R Code Script:**

```
plot(dataset$StudyHours, dataset$Marks, main = "Study Hours vs Marks",
     xlab = "Study Hours", ylab = "Marks", pch = 19, col = "darkblue")

model <- lm(Marks ~ StudyHours, data = dataset)
abline(model, col = "red", lwd = 2)
```

Q42. Predict Sales Based on Temperature Using Simple Linear Regression in R in Power BI. Estimate the regression Equation and calculate the summary statistics for both variables.

Temperature (°C)	Ice Cream Sales (1000's)
20	50
22	55
24	60
26	58
28	70
30	72
32	65
34	75
36	90
38	85

Answer:

- **Where to Write and Execute Code:**
  1. Add **R** visual to report canvas.
  2. Drag `Temperature` and `IceCreamSales` into **Values**.
  3. Paste code into **R script editor** and click **Run**.
- **R Code Script:**

```
print("Temperature Summary:")
print(summary(dataset$Temperature))

print("Sales Summary:")
print(summary(dataset$IceCreamSales))

model <- lm(IceCreamSales ~ Temperature, data = dataset)
print("Model Coefficients:")
print(summary(model))

plot(dataset$Temperature, dataset$IceCreamSales,
     main = "Temperature vs Ice Cream Sales",
     xlab = "Temperature (°C)", ylab = "Sales (Thousands)", pch = 19, col = "coral")
abline(model, col = "blue", lwd = 2)
```

Q43. Create a Dashboard combining Power BI visuals and Python/R charts.

Answer:

- **Step-by-Step Guide:**
  1. Create native Power BI visuals (Card visuals, Slicers).
  2. Embed Python or R visual tiles on the same canvas using **Py** or **R** icons in **Visualizations** pane.
  3. Drag relevant fields into **Values** field bucket for Python/R visuals.
  4. Arrange layout so native cross-filtering dynamically drives data passed into Python/R scripts.

Q44. Export Tableau Workbooks (.twbx) and Power BI Reports (.pbix).

Answer:

- **Step-by-Step Guide:**
  - **Tableau:** Go to **File** → **Save As** → Select format: **Tableau Packaged Workbook (*.twbx)** to save all data sources and sheets together.
  - **Power BI:** Go to **File** → **Save As** → Save as **Power BI Desktop File (*.pbix)**.